

# Human Computer Interaction



Lecture 1:  
Introduction

# Outlines:

- Introduction.
- HCI in our daily life.
- Why HCI matters.
- Relation between HCI and other fields.
- HCI main components.

# What Is Human-Computer Interaction?

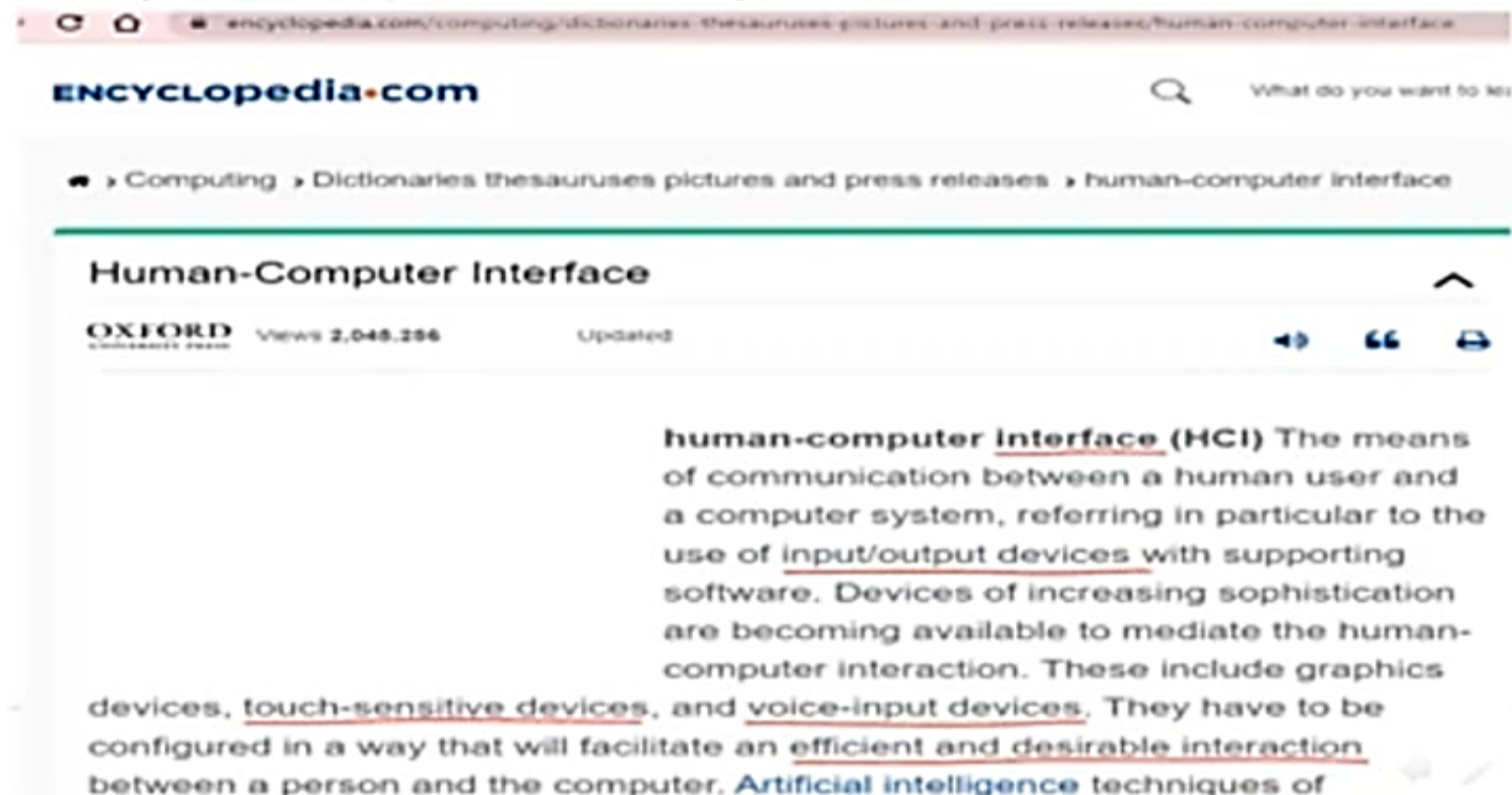
## **Introduction to Human Computer Interaction (HCI)**

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- **Human-Computer Interaction (HCI)** is a multidisciplinary field that focuses on the design, evaluation, and implementation of interactive computing systems for human use.
- It also studies the major phenomena surrounding these systems—how people interact with computers, how systems respond, and how interfaces can be improved for better usability.

# HCI

- ▶ Is the I in the abbreviation for 'Interaction' or 'Interfaces'?
- ▶ Personally, I used to think (or I was taught) it is 'Interfaces'
- ▶ Possibly, the abbreviation adapted to the better word



The screenshot shows a web browser displaying the Oxford Encyclopedia entry for 'Human-Computer Interface'. The browser's address bar shows the URL: [encyclopedia.com/computing/dictionaries-thesauruses-pictures-and-press-releases/human-computer-interface](https://www.encyclopedia.com/computing/dictionaries-thesauruses-pictures-and-press-releases/human-computer-interface). The page header includes the 'ENCYCLOPEDIA.com' logo and a search bar. Below the header, a breadcrumb trail reads: 'Computing > Dictionaries thesauruses pictures and press releases > human-computer interface'. The main title of the article is 'Human-Computer Interface', with the Oxford logo and 'Views 2,048,296' and 'Updated' information below it. The definition of HCI is provided in a text block, with several terms underlined: 'input/output devices', 'touch-sensitive devices', 'voice-input devices', and 'efficient and desirable interaction'. The text concludes with 'Artificial intelligence techniques of'.

encyclopedia.com/computing/dictionaries-thesauruses-pictures-and-press-releases/human-computer-interface

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## Human-Computer Interface

OXFORD  
UNIVERSITY PRESS

Views 2,048,296

Updated

human-computer interface (HCI) The means of communication between a human user and a computer system, referring in particular to the use of input/output devices with supporting software. Devices of increasing sophistication are becoming available to mediate the human-computer interaction. These include graphics devices, touch-sensitive devices, and voice-input devices. They have to be configured in a way that will facilitate an efficient and desirable interaction between a person and the computer. **Artificial intelligence** techniques of

# Examples of HCI in Daily Life

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- Smartphone interfaces
- ATM machines
- Car dashboards
- E-learning platforms
- Voice assistants (Siri, Alexa)
- Websites and mobile applications

Every button, menu, icon, and gesture you use is influenced by HCI principles.

# Why HCI Matters

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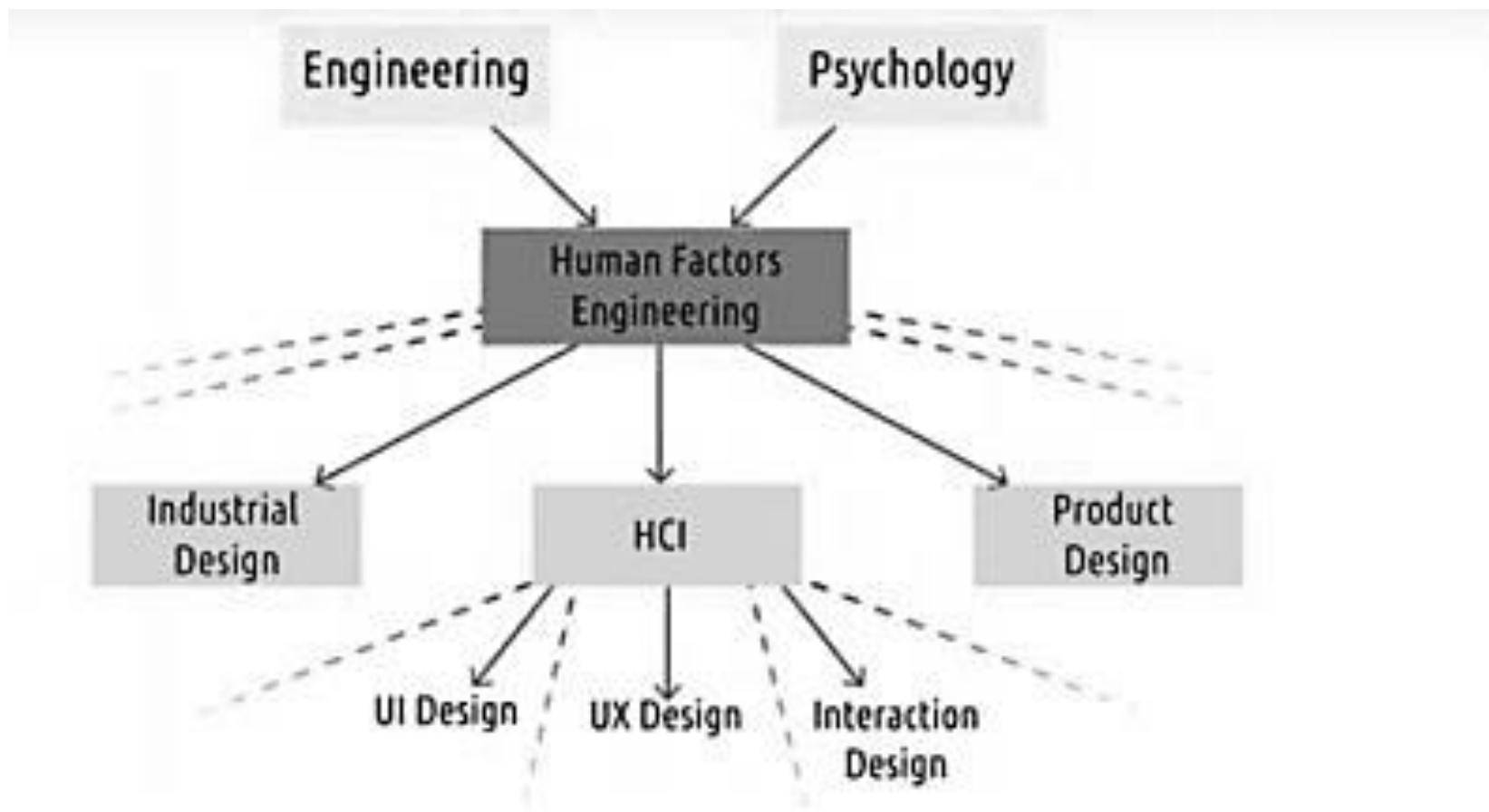
Because modern life depends heavily on digital systems—mobile apps, websites, ATMs, dashboards, smart devices—HCI ensures that:

- Users can **interact with technology easily**.
- Systems **reduce errors** and confusion.
- Interfaces are accessible to all, including people with disabilities.
- Technology becomes **human-centered**, not machine-centered.

Good HCI leads to:

- Better productivity
- Better user satisfaction
- Lower training and support costs

# HCI in a broader hierarchy of fields



## **UX Design:    User Experience Design**

Is the process of designing products or systems to provide a good, easy, and satisfying experience for users.

## **UI Design:    User Interface Design**

Is the process of designing the visual interface of a system, including elements such as buttons, colours, fonts, icons, and layouts.

## **Interaction Design**

Is the process of designing how users interact with a system and how the system responds to their actions.

| UX Design — User Experience Design                                                                                                                                                                                                                                  | UI Design — User Interface Design                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>It focuses on:</b></p> <p>Is the application easy to use?</p> <p>Can users achieve their goals easily?</p> <p>Are the steps clear and logical?</p> <p>Can users find what they need easily?</p> <p>Does the user feel comfortable while using the system?</p> | <p>Menu</p> <p>Font size and color</p> <p>Buttons</p> <p>Layout</p> <p>icons</p>                                                                  |
| <p><b>Example:</b></p> <p>In a flight booking application, UX Design focuses on making the process of searching, selecting, and booking a flight easy and clear.</p>                                                                                                | <p><b>Example:</b></p> <p>In a flight booking application, UI Design focuses on the colour, size, shape, and position of the Book Now button.</p> |

## Principles of human-computer interaction design



# HCI main component:



## **Users**

People interacting with technology



## **Tasks**

Activities users aim to complete



## **Tools**

Devices and software enabling tasks



## **Interfaces**

Connection between users and tools



## **Context**

Environment where interaction occurs

## Users:

Since the primary purpose of these technologies is to improve the lives of human beings, researchers spend quite a lot of time understanding the needs and expectations of the target audience.

Understanding the needs of the target audience helps in creating a user-centered design, ensuring usefulness and usability.

Tasks:

where the designers create various tasks and scenarios to understand the user journey.

Tools:

No technology, artifact, or interface can be created in the absence of appropriate tools. the tools need to be robust and flexible enough to allow for iterations.

## Interfaces

Helping the users navigate the design in a useful and effective manner as they try to accomplish their goals.

## Context

The situation or environment in which the user interacts with technology.

How does the environment affect the way the user interacts with the system?

# Conclusion

- HCI focuses on the interaction between people and computers.
- It aims to make systems easy, useful, efficient, and satisfying to use.
- HCI is important because good design improves the user experience and reduces errors.
- HCI is applied in many areas, such as mobile applications, websites, healthcare systems, education, and automotive systems.